

Fuzzy Fingerprinting Encoder Pre-trained Language Models for Emotion Recognition in Conversations: Human Assessment and Validity Study

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Abstract

In Emotion Recognition in Conversations (ERC), model decisions should align with nuanced human perception and ideally provide insights on the classification process. Standard encoder pre-trained language models (PLMs) are the state-of-the-art at these tasks but offer little insight into why a certain prediction is made. This is especially problematic in imbalanced datasets, where most utterances are labeled as neutral, making these models frequently misclassify minority emotions as the majority neutral class. To tackle this issue, we introduced a novel, interpretable approach to ERC by combining PLMs with Fuzzy Fingerprints (FFPs). FFP provide class-specific prototypes that reflect the characteristic class activation patterns in the PLM’s latent space. They are derived by ranking and fuzzifying the activations of the pooled conversational context-dependent embeddings across training instances for each emotion. At inference time, each input utterance is similarly fuzzy fingerprinted and matched to the emotion prototypes using a fuzzy similarity function based on the aggregation of the intersection of the fuzzy sets that define each FFP. Experimental results show that FFP integration reduces overclassification into the neutral class and human evaluation further supports the adequacy of FFP predictions. Our proposed method thus bridges the gap between deep neural inference and human perception, performing at state-of-the-art level while simultaneously offering valuable insights into the classification procedure.

Keywords: Fuzzy Fingerprints, Pre-trained Language Models, Interpretability, Imbalanced, Emotion Recognition in Conversations

1 Introduction

Emotion Recognition in Conversations (ERC) is a challenging task in which the richness and ambiguity of human emotional expression require models that are simultaneously highly accurate

and interpretable. ERC often operates in domains where human trust and psychological validity are key, such as empathetic dialogue agents, mental health monitoring, and therapeutic tools. Therefore, stakeholders must be able to understand the

rationale behind a system’s classification of emotional states. However, most state-of-the-art ERC systems rely on deep neural architectures, particularly transformer-based models, that operate as so-called black boxes.

Furthermore, ERC datasets, such as DailyDialog (Li et al. 2017), show severe label imbalance, with up to 83% of utterances labeled as *Neutral*, making state-of-the-art pre-trained language model (PLM) based classifiers misclassify minority inputs into this dominant class.

To address both interpretability and performance in these settings, we proposed integrating Fuzzy Fingerprints (FFPs) with PLMs (Pereira et al. 2023b). A diagram of our approach is shown in Figure 1. This hybrid architecture does more than just mitigate class imbalance. Rather than relying solely on softmax layers or attention weights, our method creates interpretable prototypes for each emotion, enabling instance-level explanations.

With FFPs, we can answer not just “what did the model predict?” but “**why** did the model see this utterance as fear instead of sadness or neutral”, grounded in measurable similarity to a learned emotion fingerprint.

The classification insights of FFPs stem from their structure and transparency. Unlike softmax layers that entangle all weights into a single decision boundary, FFPs explicitly rank the importance of each latent feature for every class. This produces a prototype for each emotion that can be analyzed, visualized, and compared. The fuzzification process encodes degrees of relevance, differentiating features that are strongly associated with a particular emotion from those that are more ambiguous. The model does not just output a label, it allows us to state, for example, that “this test utterance is most similar to the fingerprint for class A, due to high activation in the embedding cells X, Y, and Z.” This reasoning mimics human categorization, where we judge new stimuli based on proximity to familiar prototypes.

To validate this observation, we conduct a human evaluation validity study in which we extract all instances where the FFP classifier produces a different label than the baseline model and perform an A/B test with human annotators concerning the labels generated by both models. Results indicate that the FFP classifier aligns more closely with human judgment, as also

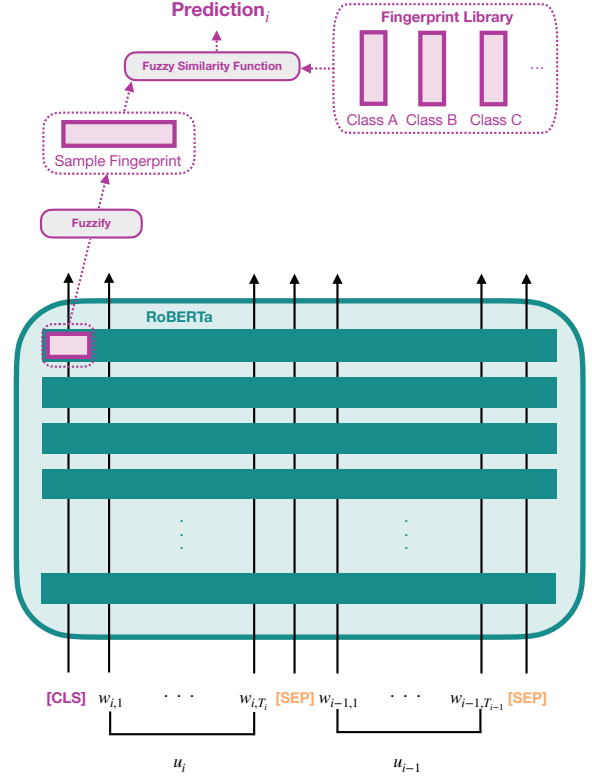


Fig. 1 Model architecture. In this example input, an utterance, u_i , and its conversational context, u_{i-1} , are fed to the encoder PLM, of which the $[CLS]$ token of the last layer goes through the Fuzzy Fingerprint module (Pereira et al. 2023b).

illustrated by the examples in Tables 9 and 10. This suggests that beyond mitigating class imbalance, our approach enhances interpretability and produces classifications that better reflect human emotional perception.

2 Related Work

2.1 Emotion Recognition in Conversations

ERC aims to assign an emotion label to each utterance in a dialogue. ERC is key for effective communication and is a crucial component in the development of responsive and socially aware systems.

A key insight in ERC is that emotions are often shaped by context — they emerge and evolve

across conversational turns. Therefore, leveraging the conversational context, i.e., the previous utterances, is crucial for accurate emotion prediction (Pereira et al. 2025). Early neural approaches (Poria et al. 2017) used LSTMs to encode the sequence of utterances, while DialogueRNN (Majumder et al. 2019) incorporated GRUs to track speaker-specific emotional states. These models explicitly considered dialogue structure, but their recurrent nature made it difficult to capture long-term dependencies.

The introduction of the Transformer (Vaswani et al. 2017) addressed this limitation through its self-attention mechanism and shorter path for gradient flow, enabling better modeling of the conversational context. Following this, encoder-based pre-trained Language Models (PLMs) such as BERT (Devlin et al. 2019), RoBERTa (Liu et al. 2019), Longformer (Beltagy et al. 2020), and DeBERTa (He et al. 2020) became standard backbones for ERC tasks.

Recent ERC approaches have built on these PLMs not only for utterance encoding but also by incorporating external reasoning or knowledge. COSMIC (Ghosal et al. 2020) uses RoBERTa to encode utterances and supplements it with commonsense information extracted from a pre-trained commonsense transformer model, along with five bi-directional GRUs. Psychological (Li et al. 2021) also uses RoBERTa and a pretrained commonsense transformer model, while introducing a graph structure of utterances processed by a graph transformer. CoMPM (Lee and Lee 2022) leverages RoBERTa with a pre-trained memory module to capture contextual information.

The recent surge in generative Large Language Models (LLMs) has brought new paradigms to classification tasks. These models, such as OpenAI’s GPT, Meta’s LLaMA, Mistral, and DeepSeek, allow for instruction-following and reasoning over multiple dialogue turns. In this setting, emotion classification is framed as a generative task: the LLM is prompted with an utterance, and possibly other relevant input, and asked to generate an appropriate label. While this strategy benefits from the general capabilities of LLMs, it also introduces new challenges: output variability, prompt sensitivity, and lack of transparency regarding the underlying reasoning process.

In contrast to all these approaches, our method leverages the representational power of PLMs

while aiming for interpretability. We encode contextual utterances using RoBERTa (Pereira et al. 2023a), and instead of using complex classification modules, we classify utterances based on their similarity to interpretable, class-specific Fuzzy Fingerprints derived from the PLM’s [CLS] output. This approach provides insights on which parts of the embeddings are most important for each emotion class.

2.2 Fuzzy Fingerprints

Fuzzy Fingerprints were introduced as a method for identifying a specific individual from a group of suspects in applications such as Mobile User Identification (Homem and Carvalho 2011b), Web User Identification (Homem and Carvalho 2011c) or Text Authorship Identification (Homem and Carvalho 2011a). In such approaches, Fuzzy Fingerprints are constructed based on the frequency of the task features. For authorship identification, a set of texts associated with a particular class is used to create the class (author) fingerprint, in which the frequency of each word in each text is used to build the fingerprint for that author. The fingerprint for a given class is the result of fuzzification and selection of the top- k features, in this case, words, based on membership values that take into account their frequencies. The collection of the fuzzy fingerprints from all classes forms the fingerprint library. Given a fingerprint library and a test instance, the instance fingerprint is generated using the same FFP creation process, and a fuzzy-based similarity function is used to find the class with the most similar fingerprint.

Conceptually, a FFP is a ranked, fuzzified set of features defining a prototype for a given class. A FFP is a fuzzy set in the discrete universe of the features, i.e., a set which contains elements with a varying degree of membership. In the original approach, the universe of features is potentially infinite and different for each class and each instance, despite all using k -sized fingerprints. Here we use a different approach (Carvalho and Ribeiro 2026) where all FFP share the same limited universe of features.

The prototypical nature of FFP, where each class is associated with a Fingerprint, enables class and instance-level explanations for classification decisions.

3 PLM Fuzzy Fingerprinting for ERC

3.1 Task Definition

Given a conversation composed of a sequence of u_i utterances with corresponding $emotion_i$ from a predefined set of emotions, the aim of ERC is to correctly assign an emotion to each utterance of the conversation. An utterance consists of a sequence of w_{it} tokens representing its T_i words, as depicted on the bottom of Figure 1.

3.2 Context-Dependent Embedding Utterance Representations

The most common approach for ERC has been to produce context-independent representations of each utterance (using PLMs), and subsequently perform contextual modeling of the obtained representations with classification modules comprising gated and graph neural networks.

In a previous work (Pereira et al. 2023a), we proposed to produce context-dependent representations of each utterance that represent not only the utterance but also a given number of previous utterances from the conversation. This context-based approach allowed us to discard the need for complex classification modules: a single fully connected linear softmax layer appended to this variation of the PLM, is enough to achieve state-of-the-art level performance. This earned us first place in a shared-task (Pereira et al. 2024).

We start by providing the PLM with the sentence we want to classify, denoted as u_i concatenated with its conversation context - a certain number of previous sentences in the conversation, $u_{i-1}, u_{i-2}, \dots$, up to u_{i-c} .

More specifically, we feed u_i into the model, with the [CLS] token before it and the [SEP] token after it, followed by the previous turns u_{i-1} up to u_{i-c} with [SEP] tokens between them, as detailed in the diagram on Figure 1.

The encoder PLM produces multiple layers of embeddings that can be used to represent the utterance and, in our proposed method, also the preceding utterances it receives as input. Each layer is composed of several tokens corresponding to the number of tokenized words in the segment. Each token is a vector with dimensions corresponding to the PLM’s hidden size.

While these embeddings are all good candidates to represent the utterance and its context, opting for all tokens across all layers can lead to an excessively memory-intensive classification layer, compromising model performance and, more importantly, interpretability. Therefore, we select the first embedding from the final layer L, specifically the [CLS] token commonly utilized for classification.

This embedding is then fed to a fully connected linear layer with softmax so that the complete model maximizes the probability of the correct labels.

For our FFP approach, we use the fully connected layer to fine-tune the PLM, but replace it afterwards with a Fuzzy Fingerprint classification module, as detailed in the next section and represented in Figure 1.

3.3 PLM Fuzzy Fingerprinting

We adapt the concept of feature frequencies to the PLM’s latent representation. Specifically, we adapt feature frequency analysis to the PLM’s final hidden state of the [CLS] classification token, which is a real-valued vector with a dimension d that is fixed for each PLM. Although the vector’s elements do not have an explicit semantic meaning, we treat the activation intensity of such elements as a proxy for their relevance within the PLM’s latent representation, and use them as features for the creation of the FFP. This allows us to extract meaningful, class-specific patterns from the opaque latent space of the PLM.

For a given emotion, the fingerprint is created by ranking and fuzzifying the d features of the [CLS] embedding that are added across all training instances of that class. Highly activated features that consistently appear across the instances are therefore assigned higher fuzzy membership scores.

3.4 FFP Creation

The procedure for creating the FFP for a given emotion is broken-down as follows:

1. **Extract representations of training utterances:** The fine-tuned context-based PLM is fed with all the training examples of the given emotion (one by one).

2. **Aggregate representations:** The d -sized vectors generated by the PLM’s output for each training example are accumulated into a d -sized fingerprint vector of real values. These aggregated embeddings represent how the PLM internally encodes that emotion.
3. **Rank elements:** The aggregated values of the fingerprint vector are replaced by their activation rank, i.e., the feature with the highest accumulated value is replaced by "1", the second most activated feature is replaced by "2", and so on. The least activated feature will obviously assume the value d . This step identifies and ranks which latent features are more relevant to each emotion.
4. **Top- k elements:** As in the original approach, the FFP only uses the top- k more relevant features for classification purposes (instead of the whole vector of d features). The remaining $(d-k)$ features are assigned a value of 0. This introduces sparsity and allows each fingerprint to be more readable and interpretable. k is an hyperparameter that is tuned on a validation set.
5. **Fuzzify elements to obtain the FFP:** Each element of the fingerprint with a value larger than 0 is assigned a fuzzy membership value based on its rank. The following fuzzifying function is used for the experiments in this work:

$$\mu_i = 1 - a \times \frac{i-1}{k}, i > 0, \forall a \in [0, 1] \quad (1)$$

in which i is the value of the feature rank, k , is the fingerprint size, and a adjusts the slope of the function (typically 1). Other functions were tested, this is the function that provided the best score.

As a result, each class will have its own FFP, a vector of d elements, of which k elements have an associated membership value $[0, 1]$. Note that the set of the top- k elements of each class is usually different. Therefore, each FFP is a fuzzy set in the discrete universe of the d features (e.g., 768 in case the used PLM is RoBERTa).

3.5 FFP Inference

After obtaining the FFP for all possible emotions (the FFP Library), classification can be

performed. Given a utterance and its context to be classified:

1. **Create the FFP of test utterance:** Use the above described procedure. In this case the FFP is created based on a single pass of the utterance and its context through the PLM (i.e. rank the activations of the embedding representation, select the top- k elements and fuzzify the resulting vector). As a result, one also obtains a fuzzy set in the discrete universe of the d features, where only the top- k most active features have a membership degree larger than zero.
2. **Compute similarity to emotion FFP:** Check the similarity of the test utterance FFP against the FFP of each emotion. The process is based on a very simple and efficient fuzzy sets operation: intersect the fuzzy sets of the utterance FFP with the fuzzy set of the Emotion FFP, and aggregate the resulting fuzzy set to obtain a crisp value. The process uses the Fuzzy Fingerprint similarity function from Equation 2,

$$\text{sim}(\Phi_E, \Phi_u) = \sum_{v=1}^d \frac{\min(\Phi_{Ev}, \Phi_{uv})}{N}, \quad (2)$$

where Φ_{xv} is the element v of FFP x . N is an optional constant used for normalization purposes.

3. **Select the emotion** with the highest similarity.

4 Experimental Setup

4.1 Dataset

DailyDialog (Li et al. 2017) is built from websites used to practice English dialogue in daily life. Table 1 resumes its main statistics. It is labeled with the six Ekman’s basic emotions (Ekman 1999), *Anger*, *Disgust*, *Fear*, *Happiness*, *Sadness* and *Surprise*, or *Neutral*. The publicly available splits of Yanran are used and the label distribution is presented in Table 2.

4.2 Training Details

From Table 2 it can be observed the DailyDialog dataset is imbalanced, not only for its dominant majority neutral class but also for the relative

Table 1 Statistics of the DailyDialog dataset

Num. dialogues	13,118
Num. turns/labels	102,879
Avg. turns per dialogue	7.9
Avg. tokens per turn	14.6

Table 2 Proportion of labels in the DailyDialog dataset

Ang	Disg	Fear	Hap	Sad	Sur	Neu
1.0%	0.3%	0.1%	12.5%	1.1%	1.8%	83.2%

Table 3 Training Parameters

Parameter	Value
Model	RoBERTa-base
Loss Function	Cross-entropy loss
Optimizer	Adam
Initial Encoder Learning Rate	1e-5
Initial Head Learning Rate	5e-5
Encoder layer-wise decay rate	0.95 per epoch
Frozen Encoder Epochs	1st
Batch Size	4
Gradient Clipping	1.0
Optimization Metric	Macro-F1
Max # of Training Epochs	10
Early Stopping # Epochs	5

imbalance between minority classes. To promote consistent performance across all classes we use the macro-F1 score for model selection.

On Table 3 we present the training parameters for obtaining the context-dependent embedding utterance representations from RoBERTa, the best performing PLM, prior to applying our FFP approaches. We also tested BERT, DeBERTa and Longformer, all in both base and large versions from the Transformers library by Hugging Face (Wolf et al. 2020).

With regards to the parameter a (Equation 1) we have observed experimentally that 0.8 is a suitable value.

4.3 Evaluation

Our reported results are an average of 5 runs corresponding to 5 distinct random seeds that are kept for a meaningful comparison of all experiments. This is motivated by the fact that results for the same experiment obtained with different

random seeds can have high variability in the macro F1-score, comparable to the improvements that we report upon state-of-the-art models. This 5 run average is in line with several ERC works.

4.4 Generative LLM Experiments

We also test LLMs, such as LLama and Mistral, specifically Llama-3.3-70B-Instruct, Llama-4-Scout-17B-16E-Instruct, Emollama-chat-13, and Mistral-Small-3.1-24B-Instruct-2503. These LLM experiments follow a zero-shot and a few-shot format, and the conversational context is also provided. The LLMs are prompted to analyze the sentences to determine the primary emotion expressed from the set of available emotions, taking into account the conversational context.

5 Results and Analysis

5.1 Fuzzy Fingerprint Size - K

We start by analysing the effect of the FFP size K on the performance (5 runs average).

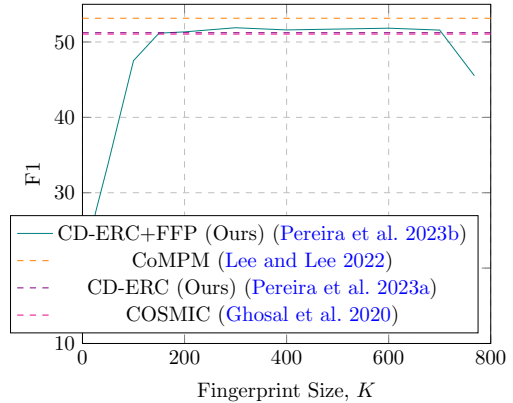


Fig. 2 Variation of the F1-score with the fingerprint size K . Other models use all RoBERTa outputs, hence having an equivalent $K=768$. Adapted from (Pereira et al. 2023b).

From the graph in Figure 2 we can see that, for K larger than 150, the performance of our proposed model is comparable to using all the 768 RoBERTa outputs (and within state-of-the-art performance level). It is therefore possible to conclude that it is not necessary to use all the RoBERTa outputs to obtain high performance,

making it possible to use a smaller and less computationally demanding model once K is decided. This also hints that it is possible to train a smaller base model for this task.

Table 4 Variation of the F1-score with the fingerprint size K

K	1	5	10	25	50	100
F1	11.67	17.22	20.04	27.21	33.78	47.52
K	150	200	300	400	600	700
F1	51.17	51.34	51.89	51.60	51.83	51.58
F1 without the Fingerprints module:						51.23

It can be observed that the peak of performance happens for $K=300$, which yields an F1-score of 51.89.

5.2 Performance on each Emotion Label

We report the F1-score on each individual emotion label with the best value for $K=300$ in Table 5 (5 runs average).

From Table 5 it can be observed that the classifiers perform better at the most represented classes in the dataset, having an F1 score of above 60 for the well-represented class *Happiness* and an even higher F1 score of around 90 for the majority *Neutral* class.

Concerning the introduction of the FFP module, it lead to an increase in F1 score in all emotion classes (except for a slight decrease in the *Anger* class), and a slight decrease in the *Neutral* class. This is in line with our observations that its introduction leads to more correct classifications with more emotional labels, as it can be seen on Table 9.

5.3 Comparison with state-of-the-art

We further compare our approach to other state-of-the-art works that also resort to PLMs. This allows for a fair comparison between approaches given that using PLMs brings great performance increases when compared to using other means of utterance feature extraction.

We compare our approach to COSMIC (Ghosal et al. 2020), RoBERTa and RoBERTa DialogueRNN, implemented by the authors of

COSMIC, and the CoMPM model (Lee and Lee 2022), all models described in Section 2. Results are displayed in table 6 (5 runs average).

We do not compare our approach to approaches that require knowledge of future utterances in the conversation (Li et al. 2021), since these are not suitable for real time ERC.

The CoMPM (Lee and Lee 2022) model has a higher performance than ours, but it resorts to two RoBERTa models, while our model leverages a single RoBERTa fine-tuned using context, and a minimalist fuzzy fingerprint classification model that can provide insights into the classification procedure.

As for our experiments with generative LLMs, these consistently underperformed in F1, yielding scores below 0.35. Interestingly, these models often reassigned neutral utterances to emotional categories, echoing the behavior of our FFP classifier.

6 Validity Studies

Interpretability in neural classification is often an afterthought, relegated to post-hoc methods that try to make sense of opaque architectures after classification. When classifying solely with PLMs or other neural architectures, it is very difficult to discern the importance of each training example towards classification of the test instance since during training the model is adapting its weights to the example/emotion pairs but one cannot accurately measure the influence of each example neither emotion in the training nor classification procedure.

Our FFP approach breaks this mold by embedding interpretability into the classification mechanism. Rather than learning hidden decision boundaries, we construct visible, interpretable fingerprints for each emotion class and match input utterances against them during the classification procedure.

6.1 FFP Visualization

Each fingerprint presented in Table 7 ($K=7$) represents a consensus activation pattern — a fuzzy set of PLM output cells ranked by their relevance to a given emotion. By distilling thousands of training examples into a compact, ranked list of salient cells, the fingerprint acts as a human-readable template for classification, allowing us

Table 5 Model performance on each individual emotion label, with and without the FFP module

Model	Ang	Disg	Fear	Hap	Sad	Sur	Neu
CD-ERC	43.51	33.22	39.44	61.12	38.43	51.50	91.42
CD-ERC+FFP	43.07	34.45	42.24	61.36	39.10	52.47	91.30

Table 6 Comparison with state-of-the-art works

	macro-F1%
RoBERTa (Ghosal et al. 2020)	48.20
RoBERTa + DialogueRNN (Ghosal et al. 2020)	49.65
COSMIC (Ghosal et al. 2020)	51.05
CD-ERC (Ours) (Pereira et al. 2023a)	51.23
CoMPM (Lee and Lee 2022)	53.15
CD-ERC+FFP (Ours) (Pereira et al. 2023b)	51.89

to determine which PLM output cells are more important for a given emotion.

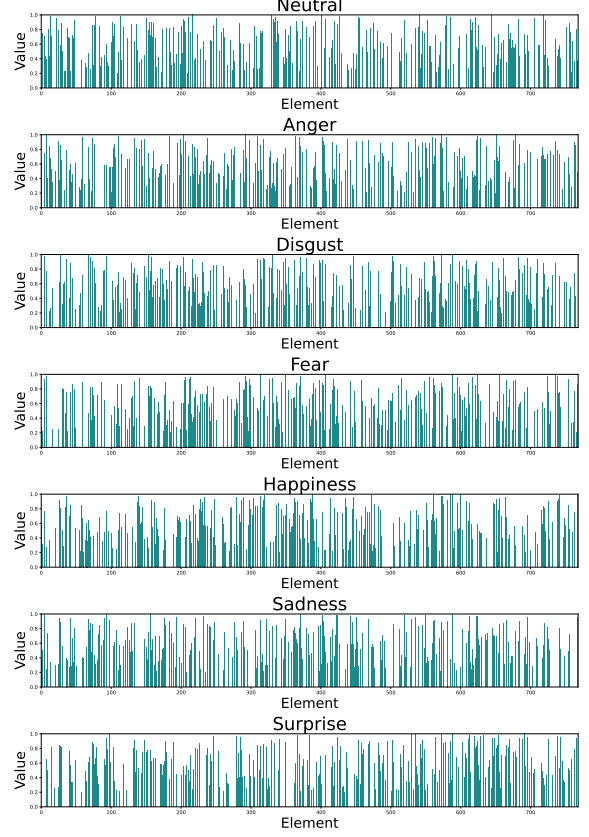
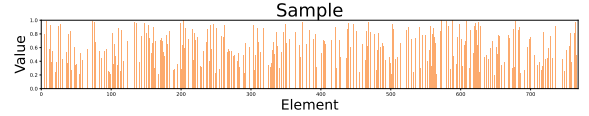
The first noticeable aspect is how output 588 is present at the fingerprints of 5 out of the 7 classes. This can be potentially leveraged to improve ERC performance using a multistage classifier, since 588 is not present in the Neutral FFP, which is the majority class and where most misclassifications end up.

In Table 8 two examples of utterance and class fingerprints, and the respective similarity results are presented.

These two examples show an interesting case of an incorrect classification and how the FFP can explain such error. The FFP of the utterance “You still have not given me those files I’ve asked you for” indicates a strong similarity to the emotion *Anger*, with many important features in common between the fingerprints. While the utterance is labelled as *Neutral*, it could certainly be argued that it has a negative meaning, revealing either an annotation error or some hidden knowledge from the annotator that cannot be extracted from the utterance. A more polite and neutral way of expressing the same message would be, for example, “Don’t forget to give me the files I’ve asked you for” (the other example on the table). Here we see that the FFP clearly indicates a *Neutral* emotion despite the sentence expressing the same message, but with a different tone.

Figure 3 ($K=300$) depicts for each emotion the unranked membership values of each element of the feature vector (RoBERTa’s 768 cells).

The fingerprint of the sample “Whatever you say !” is depicted on Figure 4.

**Fig. 3** Fingerprints of the Emotion Classes**Fig. 4** Fingerprints of the sample “Whatever you say !”

The intersection of this fingerprint with all the emotion class fingerprints is depicted on Figure 5.

It can be observed a higher density of filled element slots in the intersection with *Anger*, as well as higher membership values. This is coherent with the higher class similarity score on Table 9. There is a lower density and lower membership

Table 7 Class Fingerprints ordered by rank ($K=7$)

$FFP_{Neu} =$	$\{(217,1), (644,0.89), (541,0.77), (718,0.66), (401,0.54), (330,0.43), (426,0.31)\}$
$FFP_{Ang} =$	$\{(8,1), (679,0.89), (204,0.77), (292,0.66), (651,0.54), (573,0.43), (111,0.31)\}$
$FFP_{Dis} =$	$\{(588,1), (573,0.89), (27,0.77), (154,0.66), (331,0.54), (67,0.43), (561,0.31)\}$
$FFP_{Fear} =$	$\{(588,1), (313,0.89), (655,0.77), (406,0.66), (736,0.54), (349,0.43), (624,0.31)\}$
$FFP_{Hap} =$	$\{(588,1), (585,0.89), (388,0.77), (600,0.66), (767,0.54), (319,0.43), (741,0.31)\}$
$FFP_{Sad} =$	$\{(371,1), (588,0.89), (5,0.77), (156,0.66), (4,0.54), (93,0.43), (550,0.31)\}$
$FFP_{Sur} =$	$\{(691,1), (588,0.89), (97,0.77), (573,0.66), (530,0.54), (535,0.43), (654,0.31)\}$

Table 8 Classification examples ($K=7$), with FFP ordered by rank

Text:	You still have not given me those files I ' Ve asked you for .					
$FFP_{Sample} =$	$\{(8,1), (679,0.89), (309,0.77), (624,0.66), (292,0.54), (76,0.43), (134,0.31)\}$					
Similarity:	$Neu = 0$	$Ang = 0.35$	$Dis = 0$	$Fear = 0$	$Hap = 0$	$Sad = 0$ $Sur = 0$
Text:	Don't forget to give me the files I've asked you for					
$FFP_{Sample} =$	$\{(330,1), (644,0.89), (541,0.77), (217,0.66), (114,0.54), (426,0.43), (211,0.31)\}$					
Similarity:	$Neu = 0.44$	$Ang = 0$	$Dis = 0$	$Fear = 0$	$Hap = 0$	$Sad = 0$ $Sur = 0$

**Fig. 5** Intersection of the fingerprint of the sample “What-ever you say !” and each emotion class fingerprint.

values in the intersection with other more distant emotions, as it is the case of *Happiness*.

6.2 Case Studies

We now present examples of utterances and its class similarity scores (Equation 2).

On Table 9 we can observe examples in which the FFP classifier yielded classifications that are in line with the dataset annotations, while the standard classifier yielded incorrect *Neutral* classifications.

FFPs also help with auditing misclassifications. When the FFP classifier “disagrees” with the gold label, we can inspect the basis of its decision and potentially identify ambiguities in human annotation. On Table 10 we can observe examples in which the FFP classifier yielded classifications that are not in line with the dataset annotations, and the model without the FFP yielded *Neutral* classifications, according to the dataset. However, after reading the examples and confronting the class similarity scores, we observe that the FFP output is very reasonable and argue that the non-*Neutral* assigned emotion would make sense.

6.3 Human Evaluation Validity Study

To assess which model aligns more closely with human emotional perception, we conducted an

Table 9 FFP classification examples: labeled and classified with an emotional label by our FFP classifier and classified as neutral by our standard classifier ($K=300$, FFP ordered by rank)

Text:	Whatever you say!
Similarity:	$Neu = 0.31$ $Ang = \mathbf{0.38}$ $Dis = 0.35$ $Fear = 0.25$ $Hap = 0.18$ $Sad = 0.23$ $Sur = 0.26$
Text:	I’m a little nervous.
Similarity:	$Neu = 0.36$ $Ang = 0.24$ $Dis = 0.27$ $Fear = \mathbf{0.38}$ $Hap = 0.18$ $Sad = 0.27$ $Sur = 0.22$
Text:	This city is full of jerks
Similarity:	$Neu = 0.24$ $Ang = 0.34$ $Dis = \mathbf{0.46}$ $Fear = 0.29$ $Hap = 0.22$ $Sad = 0.26$ $Sur = 0.22$

Table 10 FFP classification examples: labeled and classified with a neutral label by our standard classifier and classified with an emotional label by our FFP classifier ($K=300$)

Text:	But it’s true.
Similarity:	$Neu = 0.23$ $Ang = \mathbf{0.39}$ $Dis = 0.29$ $Fear = 0.28$ $Hap = 0.16$ $Sad = 0.30$ $Sur = 0.27$
Text:	I wish the politicians would quit digging up dirt about each other’s past.
Similarity:	$Neu = 0.33$ $Ang = 0.32$ $Dis = \mathbf{0.37}$ $Fear = 0.25$ $Hap = 0.15$ $Sad = 0.30$ $Sur = 0.24$
Text:	No, this is not gonna happen! I... I’ve ruined everything...
Similarity:	$Neu = 0.23$ $Ang = 0.25$ $Dis = 0.27$ $Fear = 0.27$ $Hap = 0.25$ $Sad = \mathbf{0.37}$ $Sur = 0.19$

emotion annotation experiment with expert annotators, since DailyDialog was originally annotated by experts.

We targeted all test instances where the FFP classifier and the baseline model without FFP produced different emotion labels for the same utterance, resulting in collecting 69 utterances. This design isolates cases of disagreement, which are the most informative for interpretability analysis.

Each utterance was presented to three expert annotators in Computational Linguistics, all familiar with ERC tasks and discourse-level analysis, for an A/B Test.

Emotion A/B Test Guidelines: You are presented with excerpts of dyadic (two-party) dialogues. For each excerpt, your task is to choose the emotion that best describes the speaker’s feeling in the final presented utterance (the bottom one). Each final utterance is accompanied by up to 3 prior context turns to help you understand the conversation. These context turns may include the speaker of the final utterance (in blue) or the other participant (in pink). Each item will show 2 emotion options to choose from. These are randomized to avoid positional bias. For each item, choose an emotion option.

We now present an annotation example:

In this example on Figure 6, annotators favored *Anger* over *Neutral*, aligning with the label chosen by the FFP classifier. Their choice

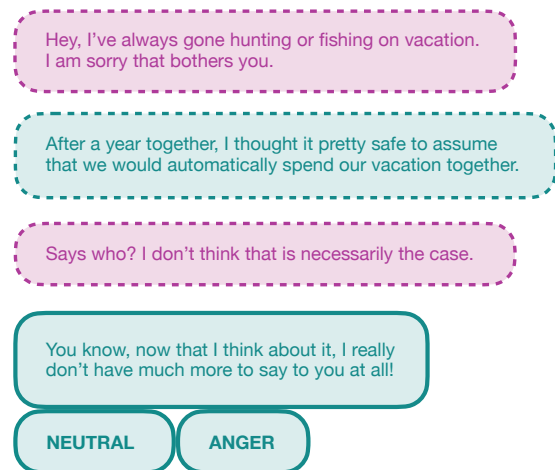


Fig. 6 Annotation example

seems the most appropriate, since the speaker is clearly not satisfied with the fact that the other interlocutor is not joining for vacation, stating that “I really don’t have much more to say to you at all!”.

Annotator preferences were towards the FFP classifier: the gold annotator preferences matched 59% of the FFP classifier labels versus 41% of the standard classifier labels. Their inter-annotator agreement, a Fleiss’ Kappa of 0.47, is very high considering that it is a typical value for emotion annotation, but in this case we isolated the most difficult and disagreement originating cases.

These results indicate that FFP classifications align more often with human perception, particularly in emotionally nuanced or context-heavy utterances. Like expert annotators, the FFP classifier shows better insight into the characteristic patterns of each class, possibly due to its prototypical nature. This capability is challenging to achieve, especially with the high number of classes at stake.

This study supports the claim that FFPs provide not only quantitative improvements (as shown in performance metrics) but also qualitative alignment with human emotional perception. In tasks where subjective interpretation and contextual nuance are critical this insight is particularly valuable.

7 Conclusion and Future Work

We have presented a text classification approach to fill the gap between interpretability and performance, adapted for the task of ERC. This approach consisted of integrating Fuzzy Fingerprints (FFPs) into a PLM-based classifier, leveraging the advantages of both. We demonstrated that it is possible to achieve state-of-the-art results while also providing insights into the classification process, since our model’s decisions are grounded in measurable similarities to interpretable class prototypes.

Results show that FFPs not only reduce the overclassification of neutral utterances, a common issue in imbalanced datasets, but also grasp emotional cues that standard neural classifiers overlook. Human evaluation validates this observation given that annotators favored the outputs of our FFP model.

Our FFP mechanism is modular and lightweight, making it suitable for integration with other types of classifiers, beyond PLM-based. We have done experiments concerning using FFPs directly on concatenated speech features for Cognitive Impairment Detection from speech (Botelho et al. 2025) in which the most difficult to detect minority class showed a great increase in F1 score.

For future work, we plan to test our approach with other types of classifiers and extend it beyond the task of ERC.

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